Ajloun National University

Faculty of Information Technology

Software requirements engineering

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ONLINE BILLS PAYMENT SYSTEM

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The online bill payment system provides the ease that enables you to pay your bills from your place and with minimal effort, and the customer is given a stamped receipt stating that the bill has been paid.

The system is linked to invoice issuers and payers, in addition to being available to those who have bank accounts or who have electronic wallets.

System Objectives:

- Facilitate and speed up the process of bill payments and other payments.

- Moving from a paper payment environment to an electronic environment to enhance the reduction of expenses.

Requirements can be extracted from more than one source:

Stakeholders, existing systems, existing documents , interviews, Brainstorming, ,Competitors and other similar systems, Interfaces with systems, Laws and Standards and Company Policies.

User Requirements:

1. user shall be able to pay the bills.

    1.1 The user shall create an account on the system if it is his first use , or access the previously created account if he is an old user.

    1.2 The user shall choose the option of "Pay the bill" then must choose the type of service, then choose the receiver and enter the bill number .

    1.3 After clicking on the "Continue" button, enter the invoice value and then click on the "Send" button.

    1.4 the user shall have bank accounts or electronic wallets. (Pre-condition)

(functional requirements)

System Requirements:

1. The system shall send a message to the user stating that the bill has been paid successfully .

Non-Functional requirements:

1. The system should be easy to use and smooth.
2. The system shall be reliable, so that its use is correct and safe.
3. the system shall ensure data security by encryption sensitive user information and transaction details to prevent unauthorized access.

- The development methodology used is the waterfall model methodology because it is a simple system that will not require many changes and modifications, and therefore after clarifying all the requirements, this methodology will be used .

Stages of the waterfall model methodology:

1. Define requirements.

2. Design: Transforming the requirements into a detailed technical design that determines how the system is built, by designing the general structure of the system, identifying components and interfaces, and determining the technologies used.

3. **Development:** Translating the design into code and developing various components and software units.

4. **Integration and testing:** This process is divided into 4 stages: unit test, component test(Assemble the developed components and test them together as a single system to verify that they are working properly & composed together correctly.) ,user test and system test.

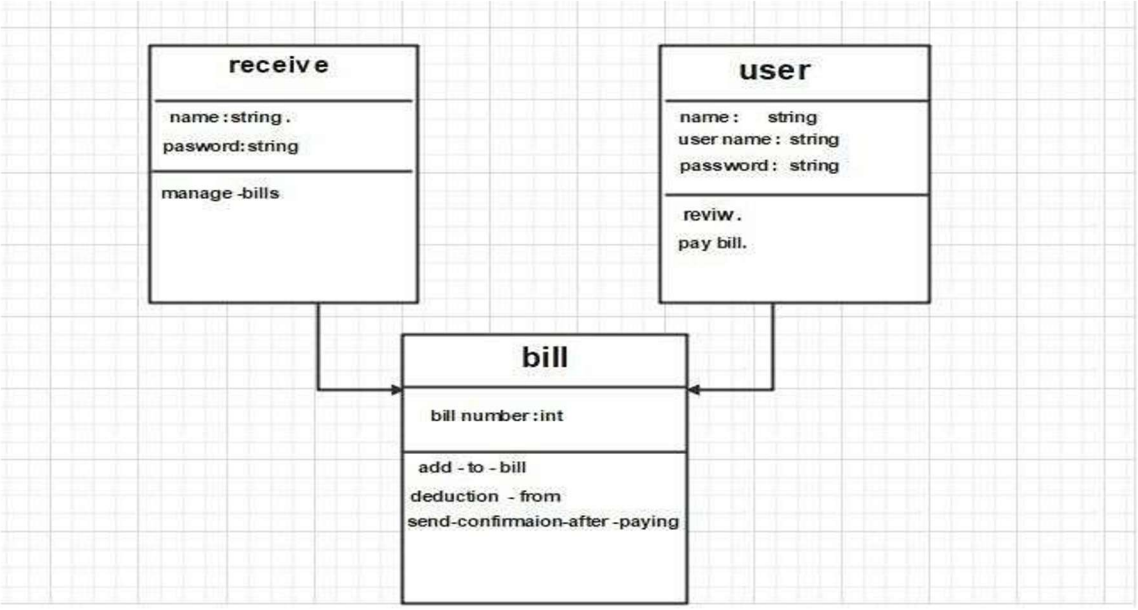
5. **Installation and deployment:** by installing the ready-made system on the operating environment.

6. **Operation and maintenance:** After installing the system, it is operated and maintained to ensure the continuity of its work and meet the requirements of users, And must take into account the times of maintenance and modification of the system and stay away from the peak time of its work.

REFERENCES

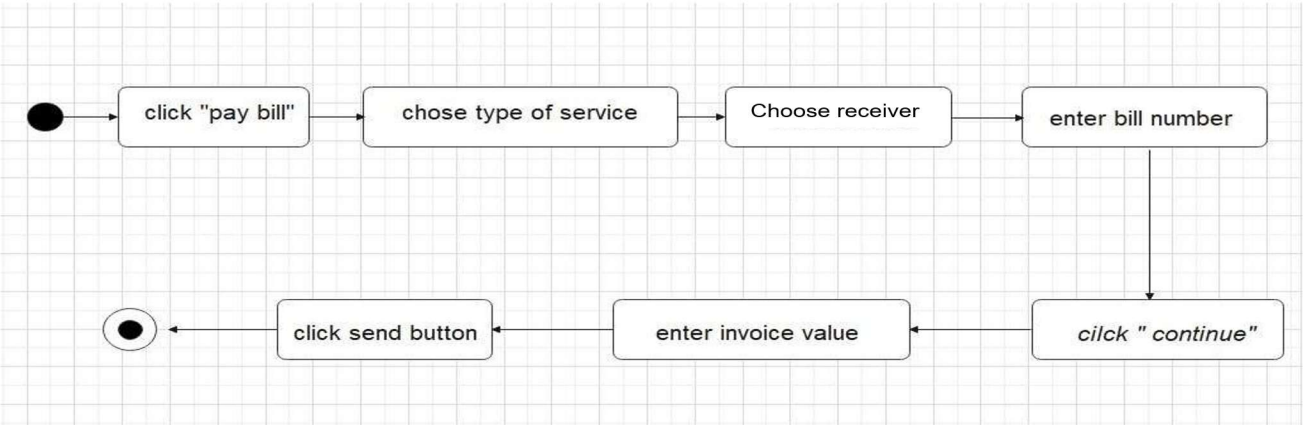
bakkah. (2024, may 12). *bakkah*. Retrieved from bakkah learning: https://bakkah.com

Class diagram



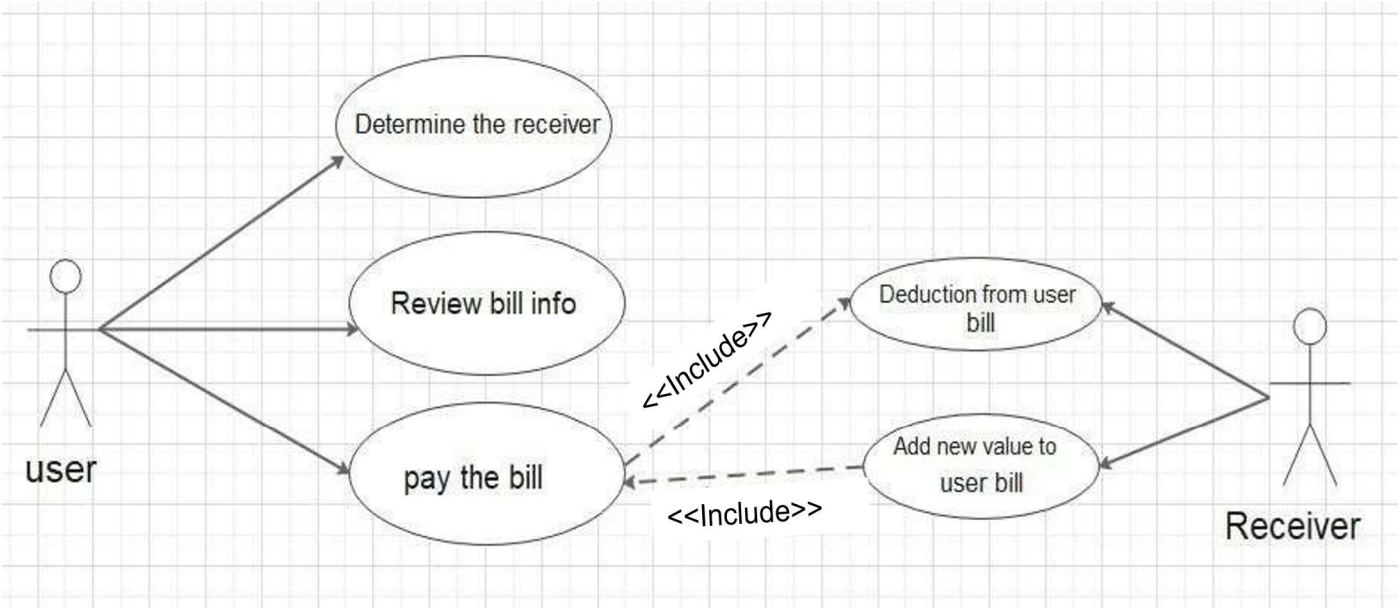
1

Activity diagram

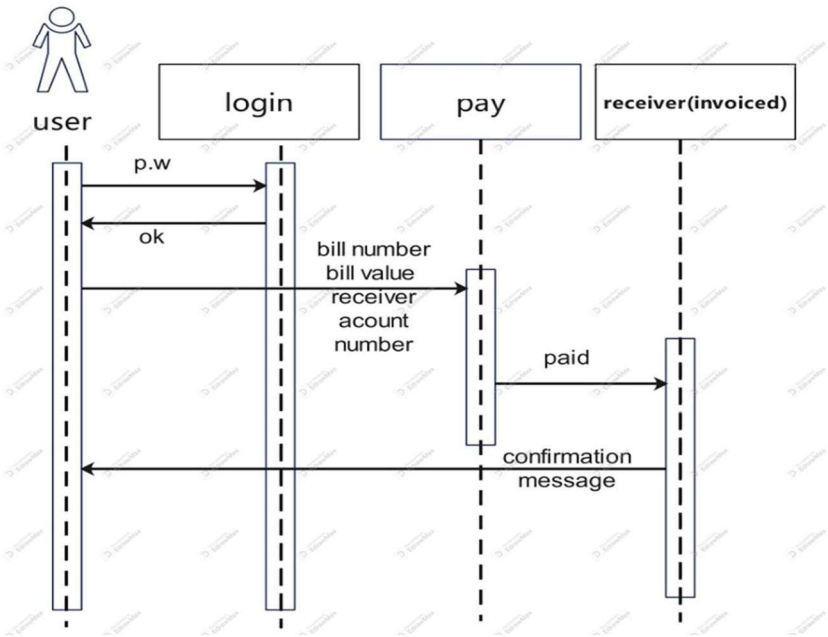


2

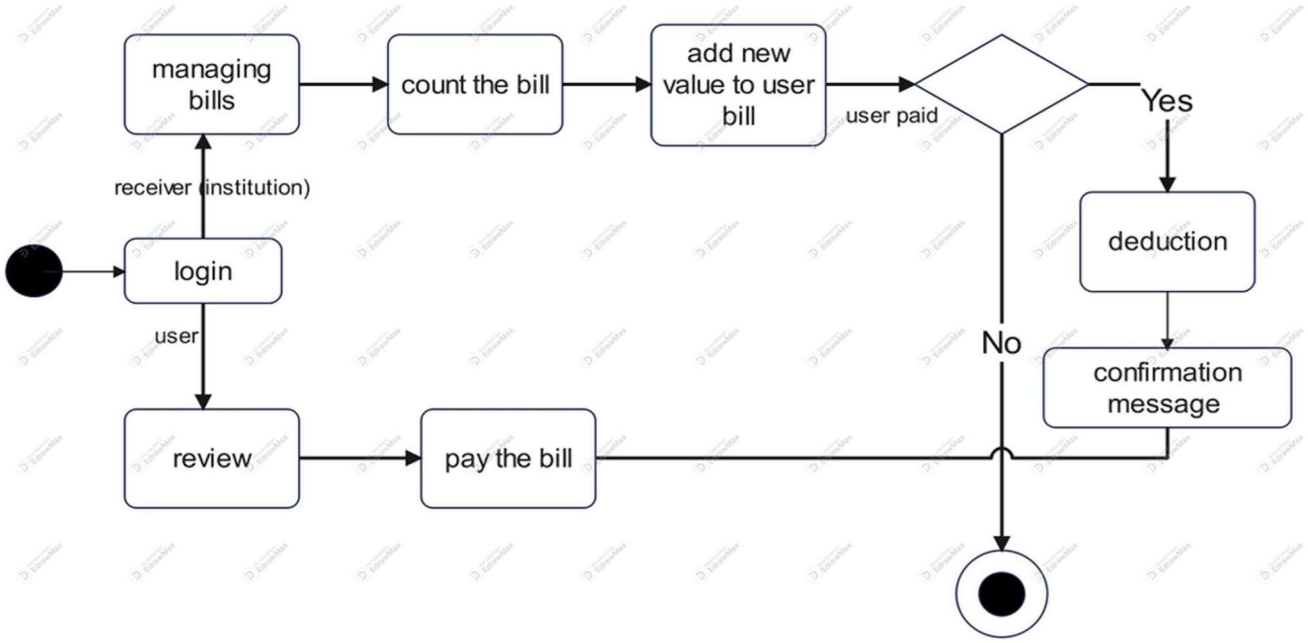
Use case diagram



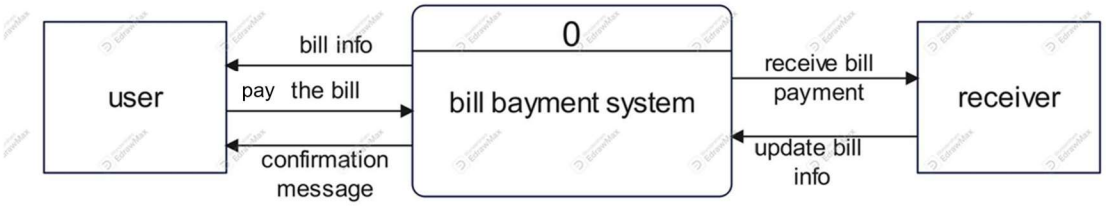
Sequence diagram



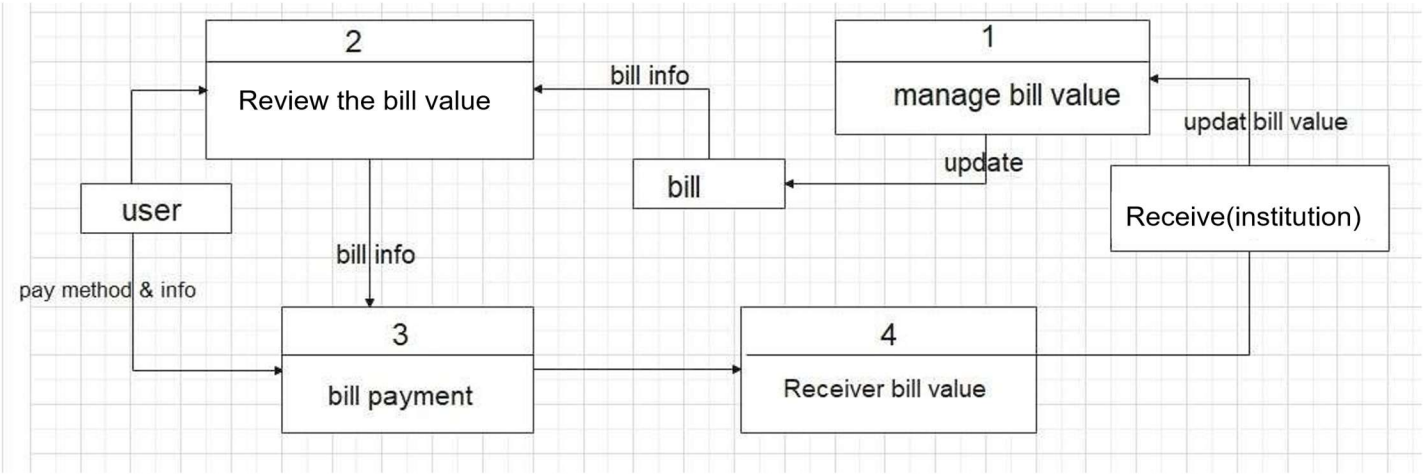
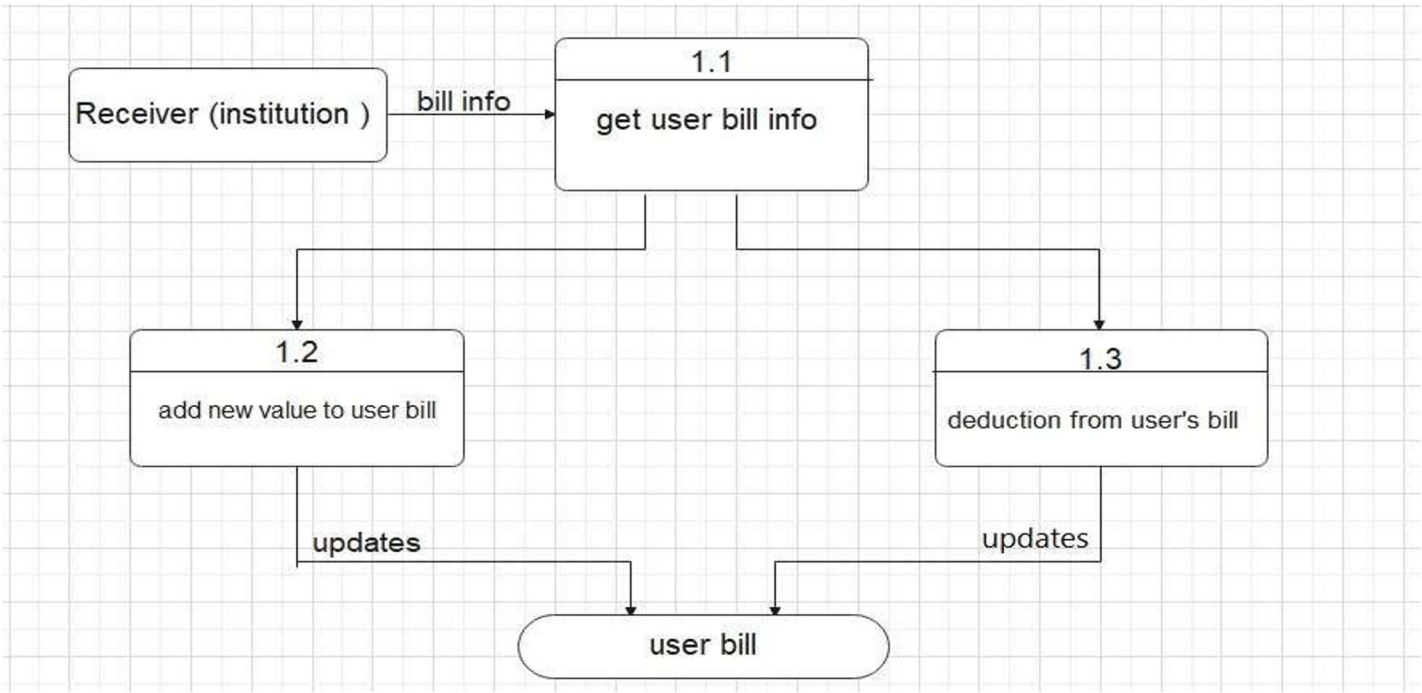
State machine diagram



Context DFD diagram



Level-0 DFD diagram



Level-1 DFD diagram